2023 Engineering Design Package (EDP)

University: The University of Illinois at Chicago	**Vehicle Name:** CST-Car
Team Captain Name: Ramiro Maldonado	**Team Captain Email:** rmaldo6@uic.edu
Faculty Supervisor: Meenesh Singh	**Supervisor Email:** mrsingh@uic.edu
Revision #: 1	**Revision Date**: 2/28/2023
***Onsite Safety Judge*:** (Virtual Only)	
***Onsite Track Length*:** (Virtual Only)	***Onsite Track Width***: (Virtual Only)

Instructions for Chem-E-Car Team:

- All Chem-E-Car Teams competing in an AIChE Chem-E-Car Competition must complete and submit and Engineering Design Package by the posted deadline.
- Failure to meet the posted deadline will result in exclusion from the competition.
- Please complete all the applicable sections of this document, and save in PDF format.
- Please rename the title of the document using the format University Name EDP.
- Additional information including Safety Data Sheets (SDS), Manufacturer's specification documents or specifications for custom-built components should be saved as another single PDF titled "University Name EDP Supplement".
- All teams will receive EDP Feedback with any suggested changes from AIChE Chem-E-Car Competition Safety Judges.
- Please use the MOC Form document to document changes made to the EDP after receiving EDP feedback from your online review.
- Each team must bring a printed copy of final EDP, EDP Supplement, EDP feedback and MOC Form in a folder or binder. This will help streamline the Onsite Safety Inspection on competition day.
- *Please review the AIChE Code of Conduct and Code of Ethic.*

For more information, visit www.aiche.org/chemecar.

Job Safety Assessment Form

Car operation, hazards, and safety: Describe your Chem-E-Car's design, intended mode of operation (propulsion system), intended mode of control (stopping mechanism), and major hazards and their control.

Describe your car's design: Our car is made with an acrylic frame of overall size 40 cm x 20 cm x 20 cm. The car has custom-built acrylic compartments for a battery case, an Arduino Uno, a relay, a photoresistor, a LED, a 5 volt motor, and a 12 v pump make up the main electrical components on our car. Our stirring mechanism is contained in a 3-D printed box where the photoresistor can read the intensity of light emitted from the LED. The stopping mechanism has a 3-D printed housing mechanism that connects the pump to the syringe that will allow us to inject solution B from a syringe into a beaker containing solution A. This box also contains a custom stir rod that uses a 5 volt motor in order to mix up the solution. A custom-made lead-acid battery powers the car. The battery has two custom-made cells that are connected in series, which powers the FA–130 electric motors to drive the car forward.

Power source (propulsion system): The propulsion system utilizes a lead-acid battery to power the motor. Lead sheets are used as the electrode for the battery cells. Four 3 in x 12 in x 3/16 in lead sheets are roughened using a meat tenderizer to increase the surface area. Those are then cut in half resulting in eight plates total. The plates are then cut 2.75 in from the bottom with a width of 1 cm and bent upwards to create a tab for the electrodes. The battery case and lid are constructed from acrylic. Polyethylene separators from old car batteries are cut to fit the size of our lead plates. The lead oxide plates are placed into the separators to prevent the positive and negative plates from touching each other inside our battery. The battery consists of two cells with four plates each, for a total of 8 lead plates. The lead and lead oxide plates are arranged one by one. The two positive and two negative plates in each cell are connected in parallel, and the two cells are connected in series. A 33% sulfuric acid solution is used as the electrolyte for the battery. The following reactions take place at the negative and positive plates respectively:

Negative Plate Reaction: $Pb(s) + HSO_4^-(aq) \rightarrow PbSO_4(s) + H^+(aq) + 2e^-$
Positive Plate Reaction: $PbO_2(s) + HSO_4^-(aq) + 3H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

Each cell can produce approximately 2.1-2.2 V and are connected in series, making this battery a 4.4 V lead-acid battery.

Auxiliary processes including synthesized fuel production, battery charging, hydrogen production, etc. (if applicable):

Stopping mechanism: The stopping mechanism is based on an iodine clock reaction. The reaction involves preparing two solutions: one consisting of acids, and one consisting of salts. The acid solution is glacial acetic acid with 6% hydrogen peroxide diluted with water. The salt solution is potassium iodide, sodium thiosulfate, sodium acetate, and soluble starch in water. The salt solution is stored in a container and placed in a beaker in a dark box between a LED that faces a photoresistor. The acid solution is injected into the salt solution via a syringe. The solution mixture is initially clear but eventually turns a dark brown as the reaction proceeds. As the reaction proceeds, the solution reaction occurs when sodium thiosulfate reduces the iodine back to iodide, and once all the thiosulfate is consumed, the iodine forms a blue-black complex with the starch to create the dark color. At a certain value of light transmittance, the sensor sends a signal to the circuit to stop the car. In order to make the change of color uniform throughout the beaker, we have built a miniature stirring mechanism. Our stirring mechanism consists of a raspberry pi fan and two small, strong magnets attached to the rotating axle of the fan

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Hazards inherent in design: The lead-acid battery uses a ~33% sulfuric acid solution, which is highly corrosive and dangerous to organic material. The acid is controlled by double containment. There are electrical components on the car like wires, batteries, etc. which can create potential shock hazards. The electrical components are all properly insulated. The stopping mechanism reaction uses a mixture of glacial acetic acid, potassium iodide, and sodium thiosulfate that can be toxic. Glacial acetic acid is flammable. It is controlled by double containment and is not near any potential ignition threats (i.e. flames).

Safety measures: Epoxy glue is used to seal all the mating parts on the battery case to further prevent any spills of the sulfuric acid. The battery is placed inside a holding unit on our car in case a spill was to occur. The holding unit is a second laser-cut acrylic box. For the stopping mechanism, the container holding the solutions is an enclosed 3-D printed box where solution B will sit in the beaker within the box and solution A will sit in the beaker attached to the pump in order to be pumped in. Within the container there is a ring that stabilizes the beaker and prevents it from shifting. The auto injector was designed so that the solution reaches only half of its volume to ensure no spill.

Please list the major design changes for your car, and how the reaction chemistry of the propulsion and stopping mechanism has been changed from the car used in the previous year:

1. **Power Source**

   N/A
2. **Stopping Mechanism**

New stopping mechanism utilizes a pump and stir system. The pump is connected to a relay which is connected to a 9 volt battery that gives it power. The stirrer is controlled within the arduino itself.

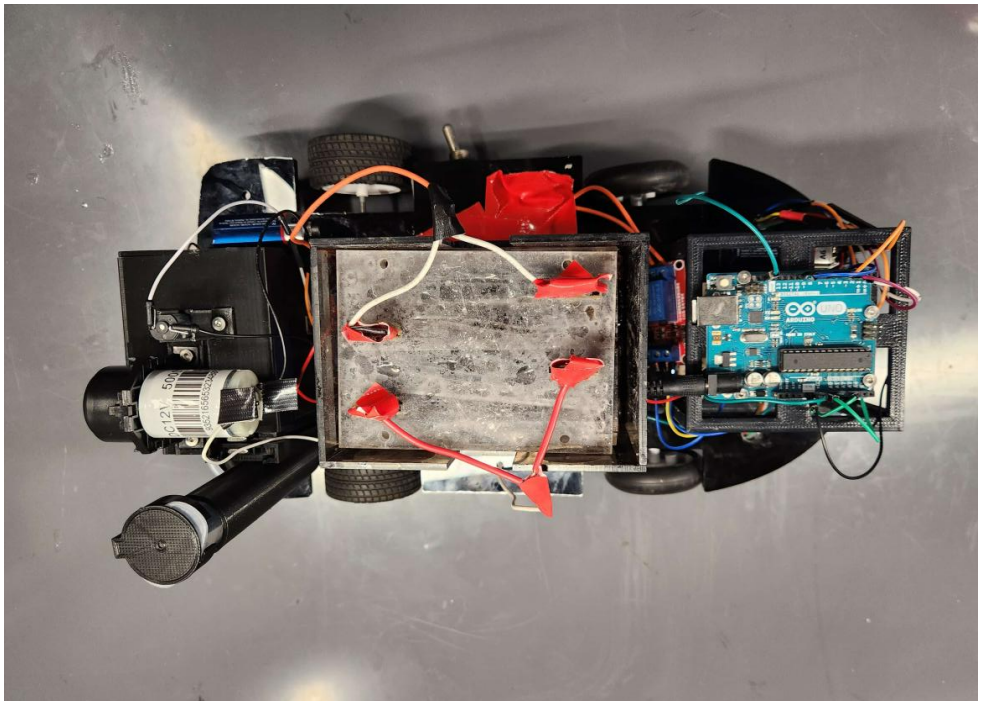
Expected Operating Conditions :
Temperature: 25℃

If your car generates pressure above 5 psig, please list the Maximum Operating Pressure (MOP) and the Maximum Allowable Working Pressure (MAWP): N/A

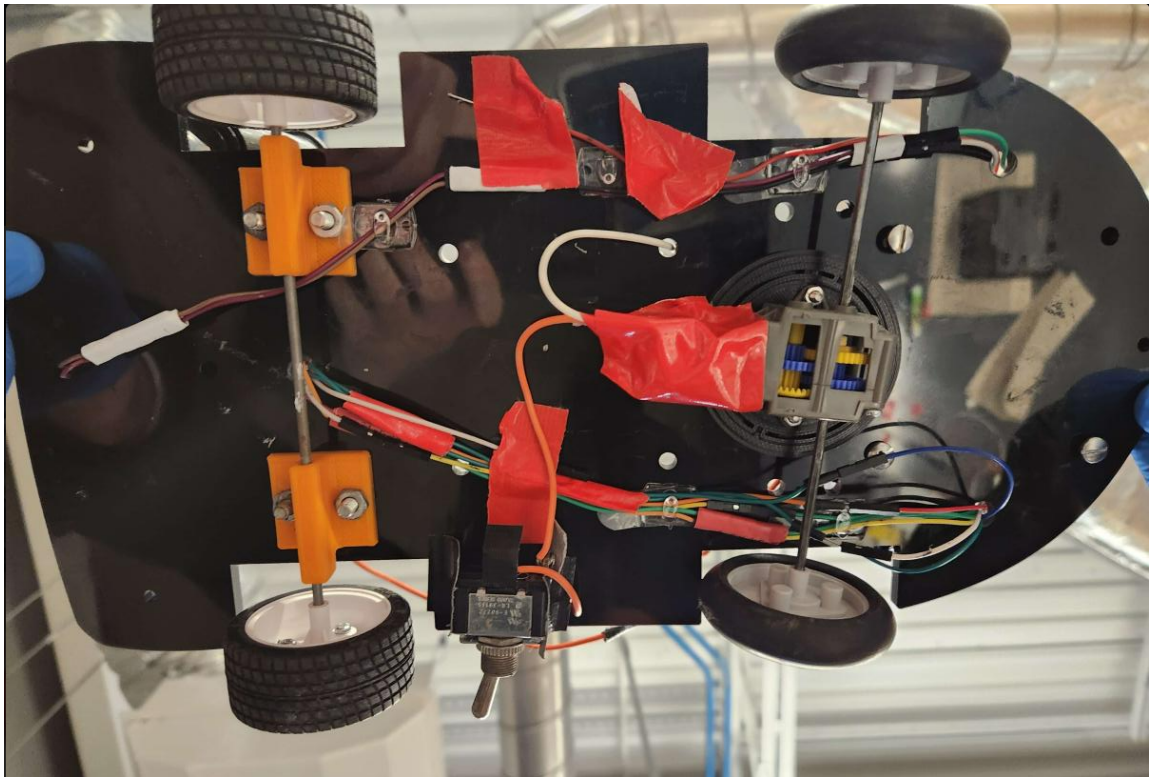
Photos of Completed Vehicle:

Please add photos of your vehicle after construction has been completed. These pictures must be current. The entire car must be visible in at least one picture. Remove the top to expose electrical controls if necessary. You must include multiple (at least 6) views of the car **(top, bottom, left, right, front and back**). Please adjust the following cells to fit *A drawing or AutoCAD document will NOT be accepted*

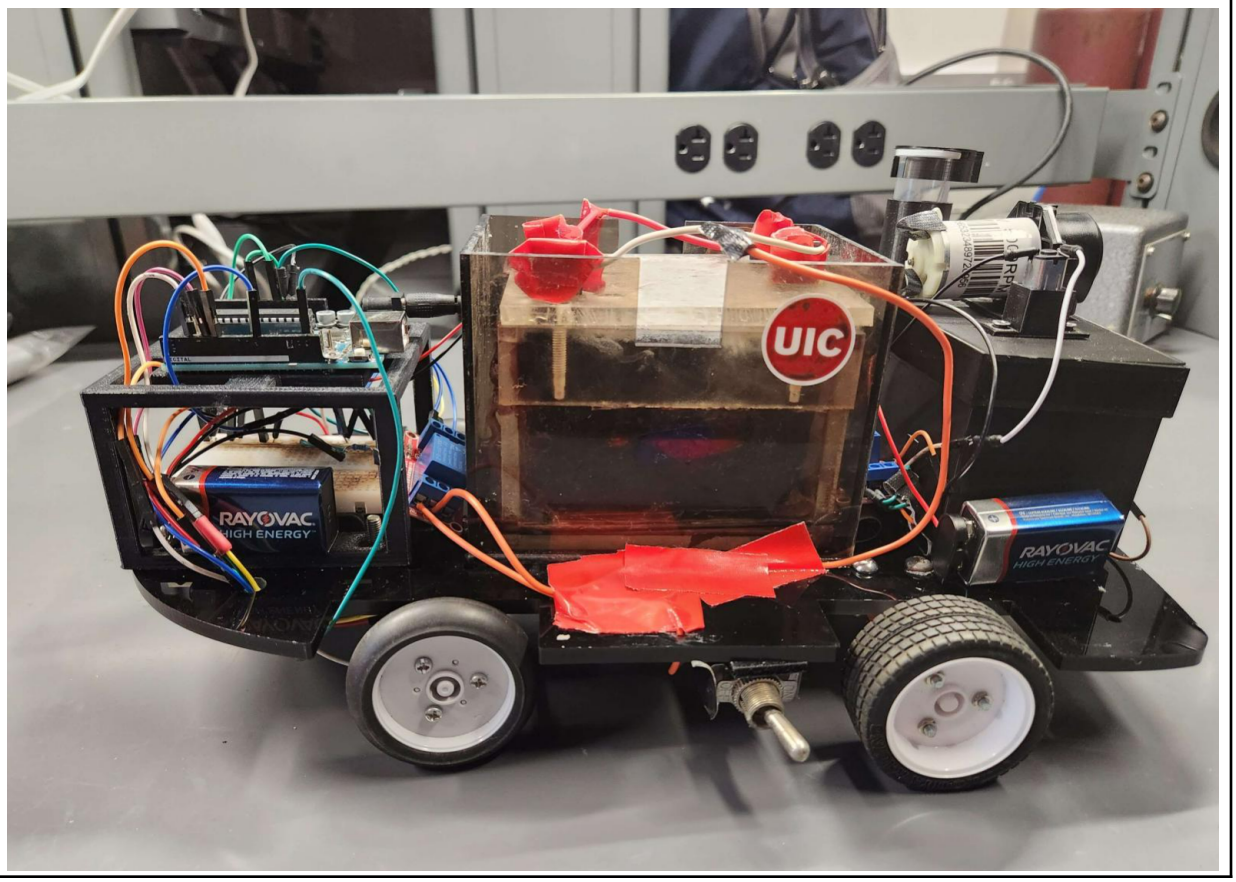
Top View



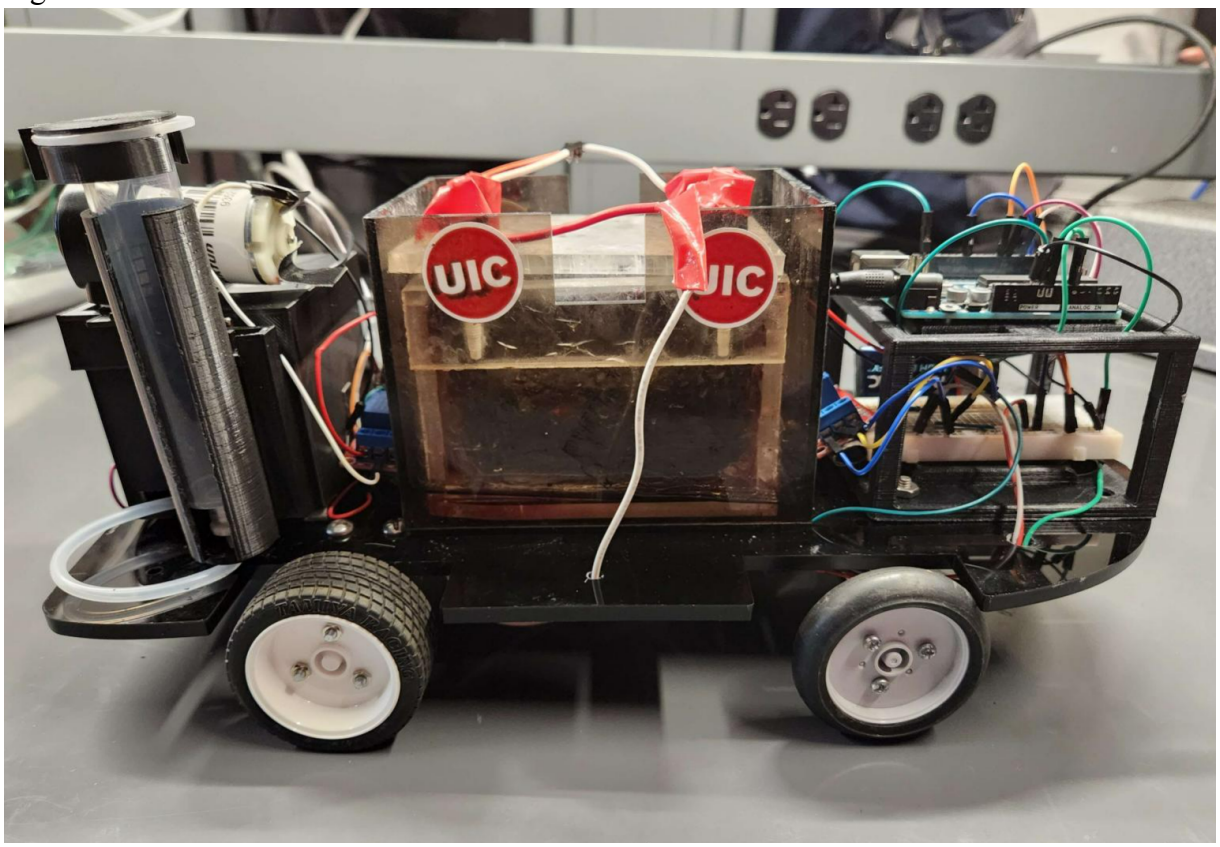
Bottom View



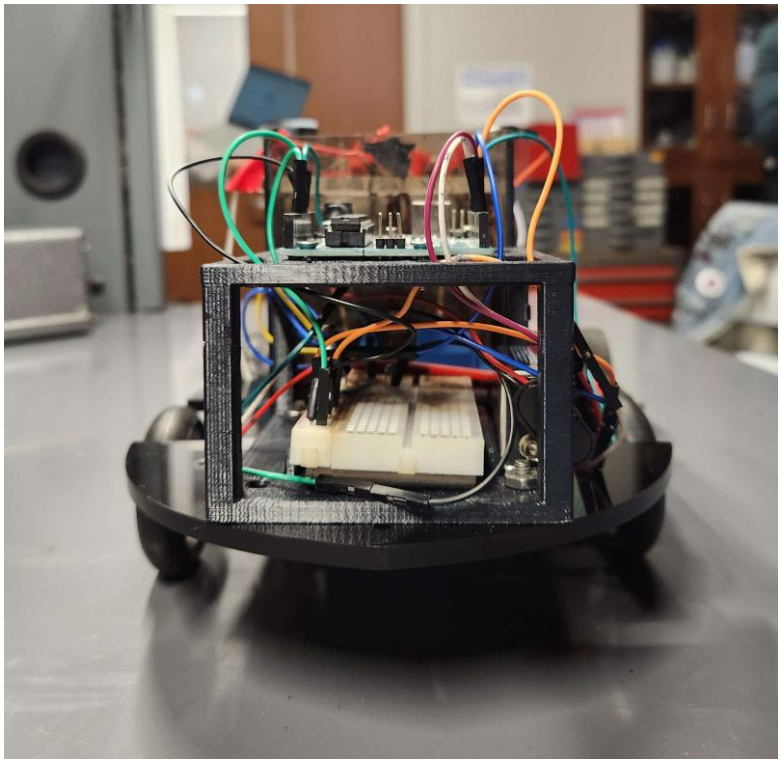
Left View



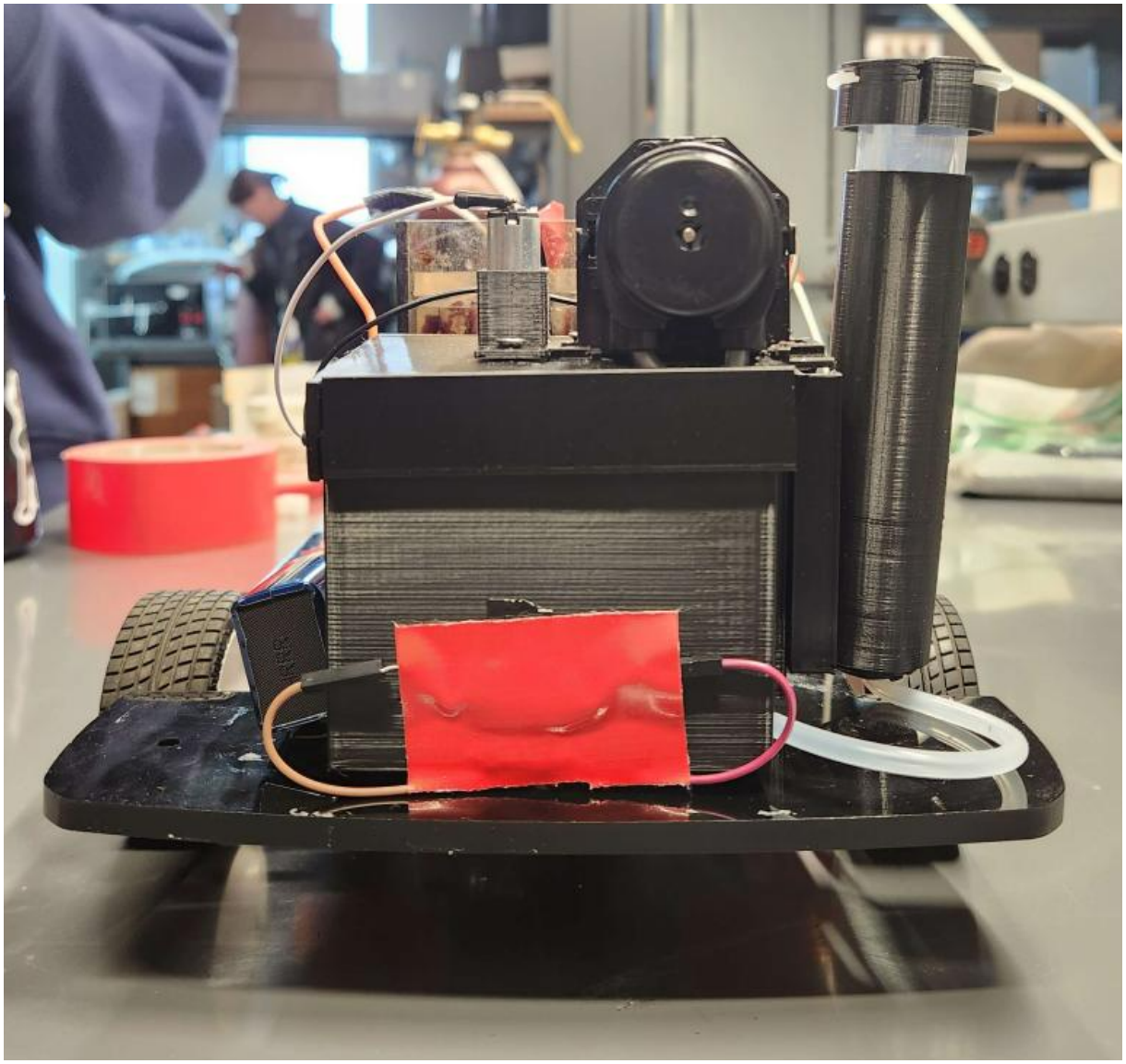
Right View



Front View



Back View



Request for Power Outlet at Competition Site (requirement for in-person competition): If your team requires a power outlet at your table at the competition, please provide reasoning here. *Please note that outlets requested to plug in laptops at your table will not be granted*.

A power outlet is necessary for the variable power supply to charge the battery.

Safety Training and Rules Certifications

University: The University of Illinois at Chicago	**Vehicle Name: CST-Car**

1. Briefly describe the propulsion system (reaction/mechanism) of your vehicle.

The propulsion system utilizes a lead acid battery to power the motor. Lead sheets will be used as the electrode for the battery cells. Polyethylene separators are used to prevent the positive and negative plates from touching. The battery case consists of two cells, each cell with four plates, for a total of 8 lead plates. 25% of the sulfuric acid solution is used as the electrolyte of the battery. The following reactions take place at the negative and positive plates respectively:

Negative Plate Reaction: $Pb(s) + HSO^-(aq) \rightarrow PbSO_4(s) + H^+(aq) + 2e^-$

Positive Plate Reaction: $PbO_2(s) + HSO_4^-(aq) + 3H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

Each cell can produce approximately 2.2 V and are connected in series, making this battery a 4.4 V lead-acid battery.

2. Briefly describe the reaction that your vehicle uses to stop at the designated finish line.

The stopping reaction is made of an iodine clock reaction which is sodium thiosulfate reacting with glacial acetic acid, sodium acetate, potassium iodide and hydrogen peroxide. The concentration of sodium thiosulfate will be increased in the reaction solution to make the reaction proceed slower, allowing the car to go farther. The concentration of sodium thiosulfate will be decreased in the reaction solution to make the reaction go faster.

3. Provide a <u>precise answer</u> to the following question: If your vehicle is 3 m short of the designated finish line on the first run, what <u>specific changes</u> will your team make to the stopping reaction to correct the shortfall?

If our car was 3 m short of the designated finish line, we would increase the volume ratio of solution A (sodium thiosulfate and other salts) to Solution B (oxidizer) used in the stopping mechanism reaction. This would prolong the reaction of triiodide ions with thiosulfate ions, therefore providing more time before the formation of the starch complex. This brown complex is responsible for the shut-down of the power source. Using the linear calibration curve, we would input the time needed for our car to go an extra 3 meters based on its constant speed of 0.5 m/s and calculate what ratio of A:B is needed.

4. Required Safety Training:

All participants must take and pass the AIChE Chem-E-Car Safety Training Course with a minimum score of 80%. For information on the Safety Training Course, please visit www.aiche.org/chemecar.
Please list the date that the required Chem-E-Car safety training that was completed by the faculty advisor:

Advisor Name	**Date**
Professor Meenesh Singh	03/12/2020

Please list the date the required Chem-E-Car safety training that was completed by each team member:

Team Member Signature	Date
Ramiro Roman	03/04/2022
Vihar Trada	03/04/2022
Mario Ocampo	03/04/2022
Karim Al Zahabi	07/25/2022
Ramiro Maldonado	07/25/2022
Emily Seriruk	11/26/2022
Andrew Russ	01/26/2023
Jocelyn Ortega	01/31/2023

Varun

5. Faculty Safety Rules Certification:
I certify that this student team has followed all of the safety and competition rules, has completed an engineering documentation package, has completed a safety review under my supervision or with an outside expert, and has at least ten hours of operating experience beyond the time required to design and assemble the car:

Faculty Advisor Name Meenesh Singh_____

Faculty Advisor Signature Meenesh Singh _____ Date: 02/04/2023___

Outside Expert Name* _____

Outside Expert Signature* _____ Date _____

******Note that this section must be signed by (1) Chem E Car Faculty advisor at a minimum. Having a signature from another outside expert is not required but recommended if you are getting support from another faculty member or safety professional on this project.*

6. Student Safety Rules Certification:
We certify that we have followed all of the safety and competition rules, have completed an engineering documentation package, have completed a safety review with our faculty supervisor or with an outside expert, and have at least ten hours of operating experience beyond the time required to design and assemble the car. We understand and agree that we will not be allowed to compete in the Chem-E-Car Competition if our completed EDP package is not resubmitted by the posted deadline:

Team Member Signature	Date
Ramiro Roman_____	03/04/2022
Vihar Trada_____	03/04/2022
Mario Ocampo_____	03/04/2022
Karim Al Zahabi_____	07/25/2022
Ramiro Maldonado _____	07/25/2022
Emily Seriruk _____	11/29/2022
Andrew Russ_____	01/26/2023
Ogechikanma Ihenacho_____	01/26/2023
Jocelyn Ortega_____	01/26/2023
Varun Maheshwari _____	01/26/2023

****Please include a copy of the Safety Training Course Certificate for each Team Member + Advisor in your EDP Supplement Document; Certificates should have a completion date of after January 1st 2021****

AIChE ACADEMY

This Certificate of Completion certifies that

Mario Ocampo

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with Professional Development Hours [PDHs] which is equivalent to Continuing Education Units [CEUs].

Date Awarded:
February 2, 2022

June C. Wispelwey, Executive Director & CEO, AIChE®

AIChE Academy
Connect | Learn | Solve

This certifies that

Ramiro Roman

Has successfully completed the eLearning course:

AIChE's ChemE Car Safety Course

Date Awarded:

September 22, 2020

Dr. Skip Rochefort
Chem-E-Car Competition Chair

AIChE ACADEMY

This certifies that

Ramiro Maldonado

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
July 25, 2022

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This Certificate of Completion certifies that

Vihar Trada

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with Professional Development Hours [PDHs] which is equivalent to Continuing Education Units [CEUs].

Date Awarded:
October 26, 2021

June C. Wispelwey, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Emily Seriruk

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
November 29, 2022

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Andrew Russ

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
January 26, 2023

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Karim Al Al Zahabi

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
July 25, 2022

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Jocelyn Alinn Ortega

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
January 31, 2023

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Varun Maheshwari

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
February 1, 2023

Darlene Schuster, Executive Director & CEO, AIChE

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AIChE ACADEMY

This certifies that

Ogechikanma Ihenacho

Has successfully completed:

ELA199 ChemE Car Safety Course

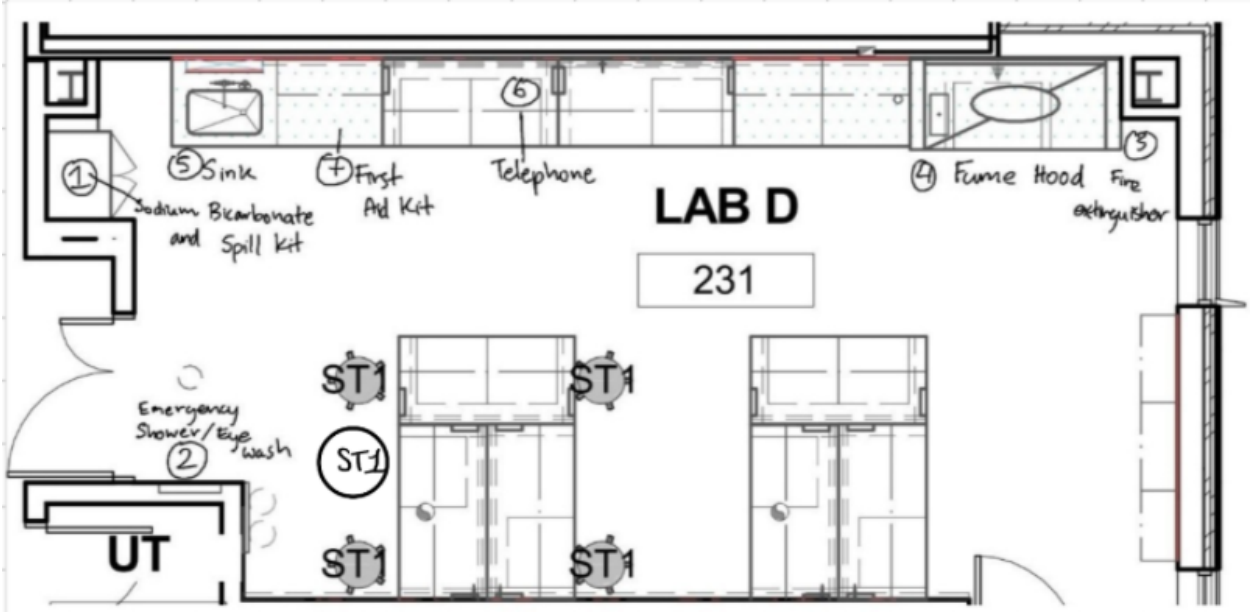
And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
February 3, 2023

Darlene Schuster, Executive Director & CEO, AIChE

Onsite Competition Fire & Safety Floor Plan

Please include a fire safety floor plan diagram of the area on campus on which you plan to run your Chem-E-Car. List the location of available fire and safety equipment as well as emergency exits on this diagram.



Hazards Analysis

This analysis is for your home institution, not the competition site. Please attach a floor diagram of the laboratory where you will be building and testing your vehicle on the following page. List the location of available safety equipment and spill response supplies on this diagram.

Expected Operating Conditions:

Temperature	Pressure
Normal: STP	Normal: STP
Minimum: STP	Minimum: STP
Maximum: STP	Maximum: STP

Personal Protective Equipment (PPE): Check all PPE worn during operation of this Chem-E-Car. Do not list these in the procedure section.

☒ Long Pants	☐ Safety Glasses	☐ Hard Hat	☐Apron
☒ Long Sleeves	☒ Splash Goggles	☒ Insulated Gloves	☐Ear Protection
☒ Non-porous Shoes	☐ Face Shield	☒ Chemical Gloves	☐Other:

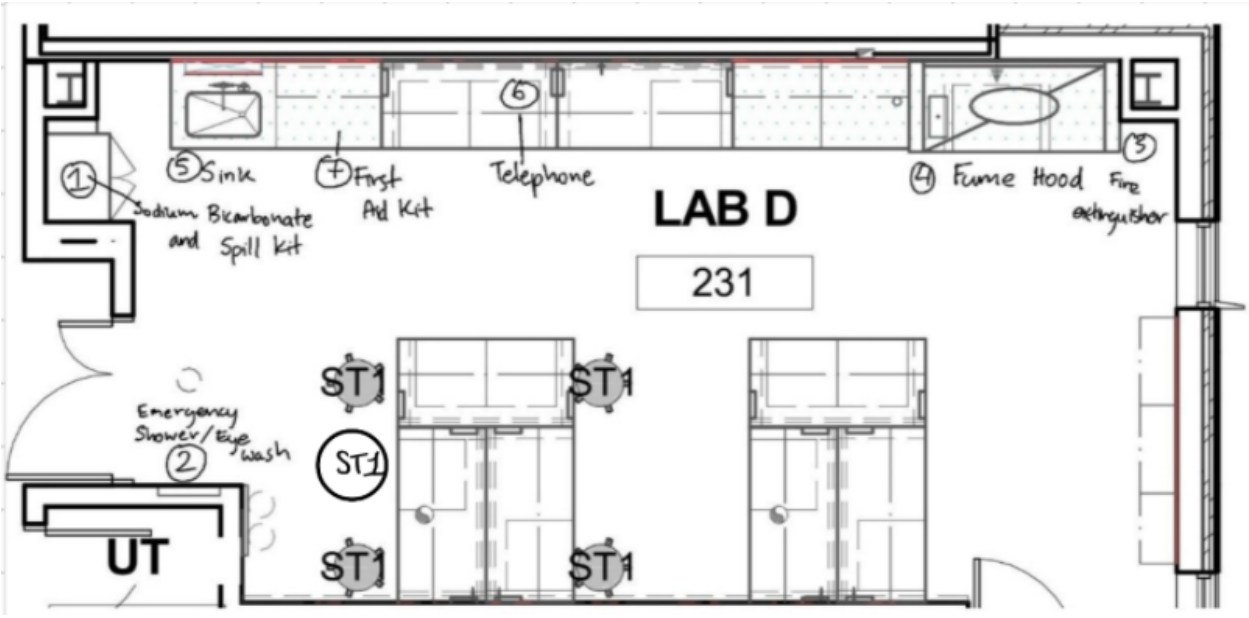
Available Safety Equipment – Provide the location of each item shown below at your home institution where your vehicle will be operated and tested. Show the location of this equipment on your provided floor plan. **Answer all questions**. If not available, type "NA" in the field.

Item	Location
Fire Extinguisher:	#3 on floor plan
Eyewash:	#2 on floor plan
Safety Shower:	#2 on floor plan
Telephone:	#6 on floor plan
First Aid Kit:	#7 on floor plan
Spill Containment	#1 on floor plan
Other: Fumehood	#4 on floor plan

Spill Response Supplies - Provide the location of each item shown below at your home institution where your vehicle will be operated and tested. Show the location of this equipment on the attached floor plan. **Answer all questions**. If not available, type "NA" in the field.

Item	Location
Spill Kit:	#1 on floor plan
Floor-Dri:	N/A
Spill Dikes:	N/A
Sodium Bicarbonate:	#1 on floor plan
Drain Plugs:	N/A
Spill Pillows:	N/A
Mercury Spill Kit:	N/A
Other:	N/A
Other:	N/A

Laboratory Floor Plan/Diagram: Please insert a floor plan diagram of the laboratory where you will be building and testing your vehicle on this page. List the location of available safety equipment and spill response supplies from the previous page on this diagram.



Vehicle Primary Hazards Checklist: Check the box in the left hand column if the hazards listed below exist on the vehicle. Then check the applicable means of control for each hazard.

Hazard (check if present)	**Control**
☐ (a) Pressure	Anything greater than 5 psig. **Must meet all requirements below:** ☐ Pressure gauge (must read to 2x max. operating pressure) ☐ Emergency relief device set to no more than 1.1 times the max. operating pressure. Relief sizing calculations must be provided. ☐ Emergency relief device in proper location. ☐ Pressure certification – see Pressure Vessel Testing Protocol ☐ Proper management system to prevent over or mis-charging. ☐ All car components exposed to pressure must be certified to operate at that pressure. Provide manufacturer's pressure specifications. ☐ No PVC, cPVC or polyethylene terephthalate (PETE or PET) plastics in pressure service or at all **Must have measurements or calculations to prove maximum operating pressure. Max allowable pressure this year is 500 psig. See ChemE car rules for more details on these requirements.**
☒ (b) Toxicity	Any chemicals with an GHS acute toxicity of 4 or lower. ☒ Doubly contained and handled properly.
☒ (c) Flammable	Any chemicals with an GHS flammability rating of 4 or lower. ☒ Doubly contained and handled properly
☒ (d) Reactive	Any chemicals with an GHS self reactive substance rating of B, C, or D. **Chemicals with a GHS self reactive substance rating of A are not allowed.** ☒ Doubly contained and handled properly.
☐ (e) Temperature	Any exposed surface greater than 150 deg. F or under 32 deg F. ☐ Insulation or barrier to prevent contact.
☐ (f) Electrical	Exposed wiring and electrically energized components are ignition, electrocution, and a shorting/fire hazard. Alligator clips and twisted wire connections are not allowed; use binding posts or banana plugs for a more secure connection. ☐ Proper electrical insulation and connections provided.
☒ (g) Mechanical	Any parts (meshing gears, belts or chains) that are pinch hazards. ☒ Guards present and adequate.
☐ (h) Oxygen	All components exposed to oxygen. These must be ☐ certified for oxygen service. ☐ thoroughly cleaned of contaminants as per instructions in rules. ☐ not used previously for other types of service.
☐ (i) Biohazards	☐ No biohazards greater than biohazard level 1 either during the design, development, preparation or competition phases of your car.

Additional Fabrication & Operation Hazard Detail Check List: Check all hazards that are likely to be encountered during your Chem-Car construction and operation. List the major source(s) of the hazard and describe how the hazard(s) will be controlled. If both construction and hazard columns are checked in an individual row, then the hazards should be identified separately for both the construction and operation.

Hazard	Present During		Control Method(s)[1]	PPE Required[1]
	Construction ?	Operation?		
Pressure	☐	☐		
Toxicity	☒	☒	Properly disposing of the toxic chemicals and properly containing the chemicals used on the car in closed containers	Safety glasses, gloves, lab coat, long pants
Flammability	☐	☐		
Reactivity/Instability	☒	☒	Sulfuric acid / iodine clock is double contained	Safety glasses, gloves, lab coat, long pants
Hot Surfaces/High Temp > 150 F (65 C)	☐	☐		
Cold Surfaces/Low Temp < 32 F (0 C)	☐	☐		
Electrical	☒	☒	Battery, Arduino. Proper electrical insulation and connections provided	
Arc welding	☐	☐		
Gas welding	☐	☐		
Lathe	☐	☐		
Milling machine	☐	☐		
Handheld power tools	☒	☐	Properly using the tools and in a safe manner	
Drill press	☐	☐		
Other mechanical hazards	☐	☐		
Paint spraying	☐	☐		
Ionizing radiation	☐	☐		
Laser radiation	☐	☐		
Asphyxiates	☐	☐		
Open flames	☐	☐		
Potential Spills	☒	☒	Safety glasses, gloves, lab coat, long pants	Safety glasses, gloves, lab coat, long pants
Biohazards:	☐	☐		
Other:	☐	☐		
Other:	☐	☐		

Chemical Information Page

Description of Chemistry/Chemical Reactions: Provide details below on any chemical reaction(s) that occur in your system. Please show the reactants involved, the stoichiometry and the heat of reaction, if available. Also list side reactions and any other reactions that may impact safety.

Iodine Clock Reaction:

(1) $3I^-(aq) + H_2O_2 + 2H^+(aq) \rightarrow I_3^-(aq) + S_4O_6^{-2}(aq)$

(2) $I_3^-(aq) + 2S_2O_3^{-2}(aq) \rightarrow 3I^-(aq) + S_4O_6^{-2}(aq)$

(3) $2I_3^-(aq) + starch \rightarrow starch - I_5 - complex + I^-(aq)$

Battery Reaction:

Negative Plate Reaction: $Pb(s) + HSO_4^-(aq) \rightarrow PbSO_4(s) + H^+(aq) + 2e^-$

Positive Plate Reaction: $PbO_2(s) + HSO_4^-(aq) + 3H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

Table 1: Pleas

Chemical Name	Chemical State (Solid, Liquid, Gas)	Concentration (MUST include units!)	Amount To Be Sent To Competition Site (for an in-person Competition only)
Sulfuric Acid (95-98%)	Liquid	18.39 mol/L	1000 mL
Water	Liquid	55.49 mol/L	1000 mL
Sodium acetate (99%)	Solid	N/A	250 g
Potassium iodide (99%)	Solid	N/A	100 g
Sodium thiosulfate (99.5%)	Solid	N/A	2 kg
Glacial Acetic Acid (99.7%)	Liquid	17.4 mol/L	500 mL
Hydrogen Peroxide (30%)	Liquid	9.8 mol/L	500 mL
Soluble Starch	Solid	N/A	500 g

e list all chemicals, concentrations and quantities that will be shipped to competition site. This is so the Host can prepare to receive, store and transport your chemicals.

Note: Please include any special storage requests that the Host should be aware of for your chemicals.

Table 2: Please list all chemicals that you expect to generate and dispose of during the competition. This should be WASTE/USED chemicals only.

Chemical Name	Concentration (Include units!)	Amount (MUST include units!)	Waste Classification (Acid, Base, Organic, Metal, Oxidizer, Other)
Sulfuric Acid	4.7 mol/L	800 mL	Acid
Deionized Water	55.49 mol/L	400 mL	Water
Hydrogen Peroxide	1.765 mol/L	125 mL	Oxidizer
Starch-I5-Complex	N/A	30 mL	Other
Sodium Acetate	0.05 mol/L	220 mL	Other

Sodium Tetrathionate	0.836 mol/L	60 mL	Other
Sodium Thiosulfate	0.759 mol/L	220 mL	Base
Glacial Acetic Acid	17.4 mol/L	7.5 mL	Acid

Chemical Hazards and Disposal

Chemical Properties and Hazards for ALL CHEMICALS, including reactants, intermediates and products.

Chemical Name	Physical State (S, L, G)	GHS Ratings*				Incompatible Chemicals List chemicals present within the laboratory	Flash Point Temp.	Flammability Limits	
		H	F	S	Sp.			LFL	UFL
Sulfuric Acid (18M)	L	3	0	2	0	N/A	N/A	N/A	N/A
Water	L	0	0	0	0	N/A	N/A	N/A	N/A
Lead	S	2	0	0	0	Strong Acids/Bases, Strong Oxidizing Agents	N/A	N/A	N/A
Lead Oxide	S	3	0	0	0	Strong oxidizing agents, Strong reducing agents, Powdered metals	N/A	N/A	N/A
Soluble Starch (Lab Grade)	S	1	0	0	0	Strong oxidizing agents, Strong reducing agents, Powdered metals	N/A	N/A	N/A
Anhydrous Sodium Acetate	S	1	1	0	0	Strong oxidizing agents, Strong reducing agents, Powdered metals	N/A	N/A	N/A
Potassium Triiodide	S	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Potassium Iodide	S	1	0	0	0	Strong oxidizing agents, Strong reducing agents, Powdered metals	N/A	N/A	N/A
Sodium Thiosulfate	S	1	0	0	0	Strong oxidizing agents, Strong acids/bases	N/A	N/A	N/A
Sodium Tetrathionate	S	2	0	0	0	Strong oxidizing agents, Strong acids/bases	N/A	N/A	N/A
Glacial Acetic Acid	L	3	2	0	0	Oxidizing agents, metals, peroxides, permanganates	104°F	N/A	N/A
Hydrogen Peroxide (30%)	L	3	1	3	OX	Strong acid/bases	N/A	N/A	N/A

*__GHS Ratings:__ **H** – Health, **F** – Flammability, **S** – Self Reactive, **Sp.** – Special

Chemical Toxicology, Regulation and Disposal: List the same chemicals that appear above, in the same order.

Chemical Name	Toxicology			Hazardous Waste Number	OSHA Regulated?	Waste Classification for Disposal	Personal Protective Equipment Specific to this Chemical
	TWA	PEL	Other				
Sulfuric Acid	1 mg/m3	1 mg/m3	-	P115	x	Acid	Gloves, Goggles, Lab Coat
Water	-	-	-	-	-	Water	Gloves, Goggles, Lab Coat
Lead	0.05 mg/m3	0.5 mg/m3	-	D008	-	-	Gloves, Goggles, Lab Coat
Lead Oxide	0.05 mg/m3	-	-	-	-	-	Gloves, Goggles, Lab Coat
Soluble Starch	15 mg/m3	15 mg/m3	-	-	x	Base	Gloves, Goggles, Lab Coat
Anhydrous Sodium Acetate	15 mg/m3	15 mg/m3	-	-	x	Base	Gloves, Goggles, Lab Coat
Potassium Triiodide	-	-	-	-	-	-	Gloves, Goggles, Lab Coat
Potassium Iodide	-	-	-	-	-	Oxidizer	Gloves, Goggles, Lab Coat
Sodium Tetrathionate	-	-	-	-	-	-	Gloves, Goggles, Lab Coat
Sodium Thiosulfate	-	-	-	-	-	Base	Gloves, Goggles, Lab Coat
Glacial Acetic Acid	-	25 mg/m3	-	P058	x	Acid	Gloves, Goggles, Lab Coat
Hydrogen Peroxide	-	1.4 mg/m3	-	-	x	Oxidizer	Gloves, Goggles, Lab Coat

Biohazards: Provide details below on any biological hazards that may occur during the design, development, preparation or competition phases of your car. Please list the biological hazards, the biohazard level, and a description of how these agents will be safety handled.

N/A

Standard / Safe Operating Procedures Page

- Provide step-by-step details for each of the sections shown below.
- Identify the hazards, the control methods and the personal protective equipment (PPE) required.
- Provide adequate detail so that the reviewers of this document will have adequate understanding of your procedure to pass judgment on the safety of your vehicle.

The **Emergency Shutdown** section should have only one or two steps required to stop your vehicle and bring it to a safe state.
The **Start-Up Procedure** section should list all the steps required to prepare your chemicals and vehicle.
The **Run Time Procedure** should describe all steps to operate your vehicle at the competition starting line.
The **Shutdown Procedure** should describe the steps normally taken to shut down your vehicle at the end of your competitive run.
The **Cleanup / Waste Disposal** section should list all the steps required to clean your vehicle of all chemicals and proper chemical disposal. Please keep in mind that at the Annual Student Conference Competition, you will not be working in a chemistry laboratory or have access to a sink. You should consider what additional supplies and steps will be needed when you are doing clean up at your table at the competition site. Please also list which waste bucket you will be using to dispose of the waste (example: Acids, Bases, Oxidizers, etc). Team must also remember that all glassware must be clearly labeled at the competition site.

Sequence of Steps	Potential Hazards	Procedure to Control Hazard	PPE or Equipment Required
Emergency Shutdown			
Walk up to car and flip on/off switch to off position. Remove battery from car.	Solutions spilling from battery/stopping mechanism.	The car will be handled with care to not spill any solution. In these unlikely scenarios. We would contain all contaminated equipment and seal in control polyethylene container	Polyethylene box, gloves, lab coat, goggles, absorbent.
Start-up Procedure			
Prepare solution A which contains 0.15g soluble starch, 1.03 g sodium acetate, 12.5 g potassium iodide, and the calculated amount of sodium thiosulfate for the required distance of the track. Dilute Solution A to 250 mL with water. Place 15 mL of solution A in a 15 mL beaker and place inside stopping mechanism. Prepare solution B with 7.5 mL of glacial acetic acid, 25 mL of 30 wt% hydrogen peroxide, and dilute to 250 mL with water. Place 15 mL of solution into syringe. Prepare a ~ 33 wt% sln of sulfuric acid by slowly mixing 330 mL of 95-98% sulfuric acid with 700 mL of DI water. (add acid to water). Fill the battery with around 250 mL of sulfuric acid and charge the battery using variable power supply at 5.3 volts. Place charged battery into double containment unit on car. Connect battery into circuit	1. A spill of the sulfuric acid or iodine clock reactants. 2. Mixing acid and water releases.	1. Prepare chemicals and move glassware with care. In case of spill, we will clear the area around the spill and use a spill kit to clean up the chemical, this includes using an absorbent over the spill. 2. The acid will be poured into the water container slowly. This will be done in a cooling bath to monitor temperature. We will pause and	Absorbent, gloves, lab coat, goggles

		startup the process as needed.	
Run Time Procedure			
Plug the Arduino to the external power supply (9V battery). Inject solution B into the stopping mechanism unit while simultaneously flipping our cars on/off switch to on. The car will proceed until the photoresistor has detected the physical change from the iodine clock reaction.	A spill of the sulfuric acid or iodine clock reactants.	Be cautious when interacting with the car. In case of spill, we will clear the area around the spill and use a spill kit to clean up the chemical, this includes using an absorbent over the spill	Absorbent, gloves, lab coat, goggles
Shutdown Procedure			
The vehicle will come to a stop. Then we will walk up to car and flip on/off switch to off position. Remove battery from car.	A spill of the sulfuric acid or iodine clock reactants.	Be cautious when interacting with the car. In case of spill, we will clear the area around the spill and use a spill kit to clean up the chemical, this includes using an absorbent over the spill	Absorbent, gloves, lab coat, goggles
Cleanup / Waste Disposal			
Between Runs Cleanup: The vehicle will come to a stop. Then we will walk up to car and flip on/off switch to off position. Remove battery from car. We will dispose of the iodine clock reaction solution in appropriate waste container. Final Cleanup: Used sulfuric Acid will be very slowly put into a waste container containing water and finally disposed of in the acid waste disposal container. The iodine clock reactants, by-products, and products will be disposed of in appropriate containers. The battery will be rinsed with deionized water which will be disposed of in the acid container.	Mixing acid and water releases heat.	The acid will be poured into the waste container slowly. Put the waste container in a cooling bath to monitor temperature and add > 2.6 L of water to the cooling bath to absorb the heat liberated from the reaction of H2SO4 and H20. We will pause and startup the process as needed.	Gloves, lab coats, googles

Equipment List

- Please list every piece of equipment on the car.
- Please include all manufacturer's specification documents or specifications for custom-built components in the EDP Supplement document.

Equipment	Manufacturer	Operating Limits: Temperature	Operating Limits: Pressure	Incompatible Materials?
Battery Case(Acrylic)	Custom-built	116°C(max)	1000kPa	ketones
Battery Case Lid(acrylic)	Custom-built	116°C(max)	1000kPa	ketones
Base of the car(acrylic)	Custom-Built	116°C(max)	1000kPA	ketones

Parafilm	Bemis	22 °C(max)	100-102 kPa	N/A
Steel Screws	Everbilt	18-22 °C	100-102 kPa	N/A
18ga Aluminum Wire	Southwire	660.32 °C(max)	1000kPa	N/A
Epoxy	JB Weld	18-22 °C	100-102 kPa	Acids/Oxidizers/Caustics
Arduino Uno	Adafruit	-30-70C	102kPa	N/A
Electronic Relay	Tongling	-30-70C	102kPa	N/A
Photoresistor	eBoot	-30-70C	102kPa	N/A
LED	Chanzon	-30-70C	102 kPa	N/A
Stopping Mechanism Case(PLA)	Custom-Built	180 °C	7700 kPa	bases
9V battery	Amazon Basics	-20-54 °C	N/A	N/A
Tamiya Gear Box	Tamiya	180 °C	7700	N/A
Tamiya Narrow Tire set	Tamiya	20 °C	10000kPa	Aromatics, mineral oil
Wires	Adafruit	660.32 °C(max)	10000kPa	oxidizers
Polyethylene Separators	N/A	N/A	N/A	N/A
Electrical Switch	N/A	75 C (max)	N/A	N/A

Note: It is highly recommended that you include a CAD or other diagram indicating where each piece of equipment is located within the design of your car. Use the same naming scheme in the diagram as in the equipment list."

Discharged Hydrogen (or other flammable chemicals) Calculations

If your car will include a small amount of hydrogen discharge, please use this space to provide calculations to prove to the reviewer that any discharged hydrogen is well below the Lower Flammable Limit (LFL)/Lower Explosive Limit (LEL) of hydrogen for the given volume of the reactor, chamber or fuel cell in which hydrogen is stored. *If your car does not use hydrogen, then this section should be left blank.*

Pressure Calculations

For all cars with pressure greater than 5psig, please complete the following in this section. The textbook "Chemical Process Safety" by Crowl and Louvar can be used as reference. Please see Appendix A of the Safety Rules for full instructions on what is required for Pressure Testing. *If your car does not generate pressure above 5 psig, you may leave this section blank.*

:

- A quantitative design basis for pressure relieving load;

- Sizing calculations for a pressure relief device;

- Test procedure and results for pressure relief.